SIVA PRAKASH T

Embedded Software Developer
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SUMMARY

Embedded Software Developer with 3+ years of experience in firmware development, real-time control systems, and cross-platform Flutter applications. Skilled in ESP32, Arduino, Teensy, and STM32 microcontrollers with expertise in UART, SPI, I2C, ESP-NOW, WiFi, BLE, 1-Wire, CAN, and RS232 communication protocols. Delivered 4 complete simulator systems (motion platform, parachute, ATGM, rocket launcher) and a sensor-driven interactive haunted house system.

SKILLS

Microcontrollers: ESP32, Arduino (Uno, Mega, Nano), Teensy, STM32, Raspberry Pi, ARM-based Microcontrollers

Communication Protocols: UART, SPI, I2C, ESP-NOW, WiFi, BLE, 1-Wire, CAN, RS232, Serial Communication

App Development: Flutter (Mobile and Desktop), Dart

Tools: Arduino IDE, PlatformIO, VS Code, Git

Core Skills: Embedded Firmware Development, Hardware Interfacing, Hardware Debugging, Motion Control, Real-Time Embedded Systems, Sensor Integration, Audio Integration, Multi-Device Synchronization, Simulator Systems

WORK EXPERIENCE

Embedded Software Developer

DB Productions, Chennai, Tamil Nadu | Aug 2022 - Present

- Developed embedded firmware for ESP32, Arduino, Teensy, and STM32 microcontroller platforms.
- Engineered real-time motion control systems requiring precise, low-latency hardware interfacing.
- Built Flutter-based desktop and mobile applications for simulator monitoring and control.
- Integrated sensors, actuators, and external hardware modules into embedded control architectures.
- Performed firmware debugging, hardware testing, and performance optimization.
- Developed low-latency real-time communication systems for synchronization across multiple devices.
- Supported system testing and performance tuning for simulator deployments.

PROJECTS

25-Seater Motion Platform Simulator

- Independently developed firmware, hardware control logic, and desktop and mobile apps for a 25-seat motion platform simulator.
- Designed real-time motion control for 25 independent seats operating simultaneously.
- Developed Flutter applications for desktop and mobile platforms.
- Developed hardware interfacing and motion synchronization logic for real-time seat control.
- Performed system debugging, firmware optimization, and hardware testing.

Parachute Simulator

- Developed embedded firmware and control logic for complete simulator operation.
- Integrated sensors and hardware modules for real-time system control and monitoring.
- Worked on communication interfaces and hardware debugging.
- Supported simulator testing and system integration.

ATGM Simulator

- Worked on embedded system integration and simulator control systems.
- Implemented communication and hardware interfacing modules.
- Supported firmware debugging and system testing.
- Developed real-time embedded control architecture for simulator operations.

Rocket Launcher Simulator

- Developed firmware and embedded control systems for simulator functionality.
- Worked on hardware interfacing and simulator operations.
- Performed debugging and real-time communication handling.
- Supported system integration and testing.

Interactive Haunted House System (Freelance Project)

- Developed sensor integration code to detect visitor presence and trigger scare events in real time.
- Implemented sound integration for synchronized audio effects triggered by sensor input.
- Built hardware interfacing and control logic for reliable, low-latency trigger response.
- Performed hardware testing and debugging to ensure consistent operation during live use.

ACHIEVEMENTS

- Built end-to-end firmware and application solutions for a 25-seat real-time motion platform simulator.
- Developed full-stack embedded solutions spanning firmware, Flutter desktop, and mobile applications.
- Achieved real-time synchronization of 25 independent motion devices using custom communication architecture.
- Delivered 4 advanced defense and entertainment simulator systems with complex embedded control logic.
- Delivered a sensor-driven haunted house system with real-time sound and scare triggers.
- Maintained high system reliability through firmware optimization and hardware debugging.

EDUCATION

Bachelor of Engineering

Jayaraj Annapackiam CSI College of Engineering, Anna University, Chennai | 2022